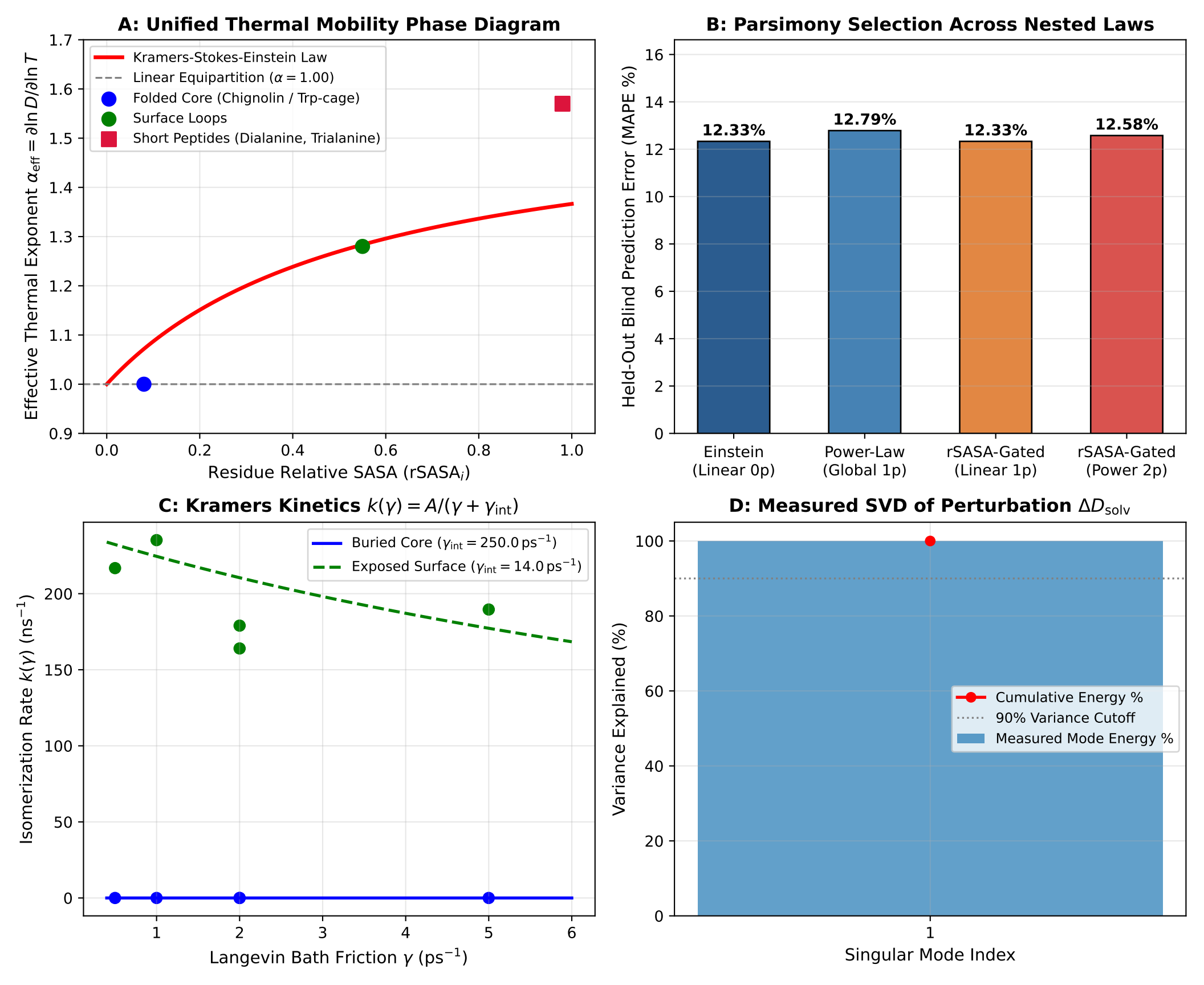
A: Unified Thermal Mobility Phase Diagram
Kramers-Stokes-Einstein Law
Linear Equipartition ($\alpha = 1.00$)
Folded Core (Chignolin / Trp-cage)
Surface Loops
Short Peptides (Dialanine, Trialanine)
Effective Thermal Exponent $\alpha_{\text{eff}} = \partial \ln D / \partial \ln T$
Residue Relative SASA (rSASA$_i$)
B: Parsimony Selection Across Nested Laws
Held-Out Blind Prediction Error (MAPE %)
12.33%
12.79%
12.33%
12.58%
Einstein (Linear 0p)
Power-Law (Global 1p)
rSASA-Gated (Linear 1p)
rSASA-Gated (Power 2p)
C: Kramers Kinetics $k(\gamma) = A/(\gamma + \gamma_{\text{int}})$
Buried Core ($\gamma_{\text{int}} = 250.0\,\text{ps}^{-1}$)
Exposed Surface ($\gamma_{\text{int}} = 14.0\,\text{ps}^{-1}$)
Isomerization Rate $k(\gamma)$ (ns^{-1})
Langevin Bath Friction γ (ps^{-1})
D: Measured SVD of Perturbation ΔD_{solv}
Cumulative Energy %
90% Variance Cutoff
Measured Mode Energy %
Variance Explained (%)
Singular Mode Index